Functional Analysis

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1. Metric spaces

Definition 1.1 (Metric space)**.** Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is a *metric* if for all $x, y, z \in X$ it satisfies $d(x,y) \geq 0$ with equality exactly when $x = y$, the symmetry $d(x,y) = d(y,x)$, and the triangle inequality $d(x,z) \leq d(x,y) + d(y,z)$. The pair (X,d) is a *metric space*.

Definition 1.2 (Cauchy sequence)**.** A sequence (x_n) in a metric space (X,d) is *Cauchy* if for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon$ for all $n, m > N$.

Definition 1.3 (Completeness)**.** A metric space (X,d) is *complete* if every Cauchy sequence in X converges to a limit lying in X.

Example 1.4 (Shortest-path metric)**.** Let $G = (V,E)$ be a connected graph and let $d(v_1, v_2)$ be the length of a shortest path between v_1 and v_2. Then d is a metric on V. Connectedness is what makes d finite.

Proposition 1.5 (Completion)**.** For every metric space (X,d) there is a complete metric space $(\overline{X}, \overline{d})$ together with a map $i\colon X \to \overline{X}$ such that $\overline{d}(i(x), i(y)) = d(x,y)$ for all $x, y \in X$ and $i(X)$ is dense in $\overline{X}$. It is unique up to isometric isomorphism: any two such spaces are related by a bijective isometry carrying one copy of X onto the other. $(\overline{X}, \overline{d})$ is the *completion* of (X,d).

Example 1.6. The space $(\mathbb{Q}, |\cdot|)$ is not complete; its completion is $(\mathbb{R}, |\cdot|)$.

Remark 1.7. It is common to suppress i and assume outright that $X \subset \overline{X}$, in which case the last condition reads: X is dense in $\overline{X}$.

2. Normed spaces and the p-norms

Definition 2.1 (Norm)**.** Let X be a vector space over $\mathbb{K}$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. A function $\|\cdot\|\colon X \to \mathbb{R}$ is a *norm* if for all $x, y \in X$ and $\lambda \in \mathbb{K}$ it satisfies $\|x\| \geq 0$ with equality exactly when $x = 0$, the homogeneity $\|\lambda x\| = |\lambda| \|x\|$, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. Then $(X, \|\cdot\|)$ is a *normed space*, and $d(x,y) = \|x - y\|$ is a metric on X.

Example 2.2.

1. $X = \mathbb{K}^n$ with the p-norm $\|x\|_p = (\sum_{j=1}^{n} |x_j|^p)^{1/p}$ for $1 \leq p < \infty$, or with $\|x\|_\infty = \max_{1 \leq j \leq n} |x_j|$.
2. $X = C[0,1]$ with the supremum norm $\|f\|_\infty = \max_{t \in [0,1]} |f(t)|$, or with the p-norm $\|f\|_p = (\int_0^1 |f|^p)^{1/p}$.
3. $X = C^1[0,1]$ with $\|f\| = \max_t |f(t)| + \max_t |f'(t)|$.
4. $\ell_p = \left\{ x = (x_n) \;\middle|\; \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}$ with $\|x\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$.
5. ℓ_∞, the bounded sequences, with $\|x\|_\infty = \sup_n |x_n|$.

Of the norm axioms, only the triangle inequality for the p-norm is not immediate. The next two theorems supply it.

Theorem 2.3 (Hölder's inequality)**.** Let $1 < p < \infty$ and let q be its *conjugate exponent*, defined by $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $x, y \in \mathbb{K}^n$,

$$\|x \cdot y\|_1 \;=\; \sum_{j=1}^{n} |x_j y_j| \;\le\; \Big(\sum_{j=1}^{n} |x_j|^p\Big)^{1/p} \Big(\sum_{j=1}^{n} |y_j|^q\Big)^{1/q} \;=\; \|x\|_p \|y\|_q, \tag{2.1}$$

where $x \cdot y = (x_1 y_1, \dots, x_n y_n)$. The inequality also holds for the endpoint pair $(p, q) = (1, \infty)$ and, by symmetry, for $(\infty, 1)$.

Remark 2.4. For $p = q = 2$ the inequality is the Cauchy–Schwarz inequality.

Proof. The endpoint case is immediate: $\sum_j |x_j y_j| \le (\max_j |y_j|) \sum_j |x_j| = \|x\|_1 \|y\|_\infty$.

Let now $1 < p < \infty$. The first step is Young's inequality: for all $a, b \ge 0$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}. \tag{2.2}$$

If $a = 0$ or $b = 0$ the left side vanishes and the right side is non-negative, so assume $a, b > 0$ and put $u_1 = a^p > 0$, $u_2 = b^q > 0$, $\alpha_1 = \frac{1}{p}$, $\alpha_2 = \frac{1}{q}$, so that $\alpha_1 + \alpha_2 = 1$. The logarithm is concave on $(0, \infty)$, since $(\log)''(t) = -1/t^2 < 0$, and concavity applied to the convex combination $\alpha_1 u_1 + \alpha_2 u_2$ gives

$$\log\big(u_1^{\alpha_1} u_2^{\alpha_2}\big) = \alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2).$$

The logarithm is increasing, so $u_1^{\alpha_1} u_2^{\alpha_2} \le \alpha_1 u_1 + \alpha_2 u_2$, which is exactly (2.2), because $u_1^{\alpha_1} u_2^{\alpha_2} = (a^p)^{1/p} (b^q)^{1/q} = ab$.

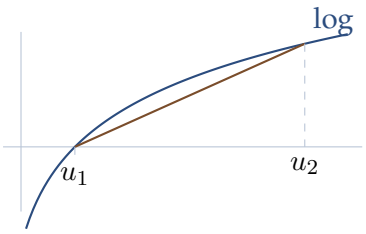


Figure 2.1. *Concavity of the logarithm: the chord over u_1 and u_2 lies below the graph, so $\alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2)$.*

The second step is a reduction by scaling. If $x = 0$ or $y = 0$ both sides of (2.1) vanish, so assume $\|x\|_p > 0$ and $\|y\|_q > 0$ and put $\tilde{x} = x/\|x\|_p$ and $\tilde{y} = y/\|y\|_q$, so that $\|\tilde{x}\|_p = \|\tilde{y}\|_q = 1$. Both sides of (2.1) are positively homogeneous of degree one in x and in y separately, so (2.1) for the pair x, y is equivalent to (2.1) for the normalised pair $\tilde{x}, \tilde{y}$. It therefore suffices to treat the case $\sum_j |x_j|^p = \sum_j |y_j|^q = 1$.

In that case, applying (2.2) with $a = |x_j|$ and $b = |y_j|$ and summing over j,

$$\sum_{j=1}^{n} |x_j y_j| \le \sum_{j=1}^{n} \Big(\frac{|x_j|^p}{p} + \frac{|y_j|^q}{q} \Big) = \frac{1}{p} + \frac{1}{q} = 1 = \|x\|_p \|y\|_q. \qquad \blacksquare$$

Theorem 2.5 (Minkowski's inequality)**.** Let $1 \le p < \infty$. Then $\|x + y\|_p \le \|x\|_p + \|y\|_p$ for all $x, y \in \mathbb{K}^n$.

Proof. For $p = 1$ the claim is the triangle inequality in $\mathbb{K}$ applied termwise. Let $1 < p < \infty$ and let q be the conjugate exponent, so that $q = \frac{p}{p-1}$ and hence $(p-1)q = p$. If $\sum_j |x_j + y_j|^p = 0$ the left side vanishes and the claim is trivial, so assume the sum is positive; being a finite sum, it is also finite. Using $|x_j + y_j| \le |x_j| + |y_j|$ and then Hölder (2.1) on each of the two resulting sums, with exponents q and p,

$$\begin{aligned}\sum_j |x_j + y_j|^p &\le \sum_j |x_j + y_j|^{p-1}|x_j| + \sum_j |x_j + y_j|^{p-1}|y_j| \\ &\le \Big(\sum_j |x_j + y_j|^{(p-1)q}\Big)^{1/q} \Big[\Big(\sum_j |x_j|^p\Big)^{1/p} + \Big(\sum_j |y_j|^p\Big)^{1/p}\Big] \\ &= \Big(\sum_j |x_j + y_j|^p\Big)^{1/q} \big(\|x\|_p + \|y\|_p\big).\end{aligned}$$

Dividing by the positive finite number $(\sum_j |x_j + y_j|^p)^{1/q}$ and using $1 - \frac{1}{q} = \frac{1}{p}$,

$$\|x + y\|_p = \Big(\sum_j |x_j + y_j|^p\Big)^{1/p} \le \|x\|_p + \|y\|_p. \qquad \blacksquare$$

2.1. Infinite sequences and functions

Both inequalities carry over from $\mathbb{K}^n$ to ℓ_p and to $C[0,1]$.

Theorem 2.6. Let $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. For all $x \in \ell_p$ and $y \in \ell_q$, $\|x \cdot y\|_1 \le \|x\|_p \|y\|_q$, and for all $f, g \in C[0,1]$,

$$\|fg\|_1 = \int_0^1 |fg| \le \Big(\int_0^1 |f|^p\Big)^{1/p} \Big(\int_0^1 |g|^q\Big)^{1/q} = \|f\|_p \|g\|_q.$$

Theorem 2.7. Let $1 \le p < \infty$. For all $x, y \in \ell_p$, $\|x+y\|_p \le \|x\|_p + \|y\|_p$, and for all $f, g \in C[0,1]$, $\|f + g\|_p \le \|f\|_p + \|g\|_p$.

2.2. The spaces $L_p[0,1]$

Definition 2.8 ($L_p[0,1]$)**.** Let $1 \le p < \infty$. On $C[0,1]$ the p-norm is a norm, but the space $(C[0,1], \|\cdot\|_p)$ is not complete. Its completion, in the sense of Proposition 1.5, is denoted $L_p[0,1]$.

Remark 2.9. The space $L_p[0,1]$ has a concrete description. Its elements are represented by those measurable functions $f\colon [0,1] \to \mathbb{K}$ for which $\int_0^1 |f|^p < \infty$, and two such functions represent the same element exactly when they agree outside a set of measure zero.

Remark 2.10. More generally, let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \le p < \infty$. Then

$$L_p(X, \mathcal{A}, \mu) = \Big\{ f\colon X \to \mathbb{K} \,\Big|\, f \text{ measurable}, \int_X |f|^p \mathrm{d}\mu < \infty \Big\}, \qquad \|f\|_p = \Big(\int_X |f|^p \mathrm{d}\mu\Big)^{1/p},$$

and $L_p(X, \mathcal{A}, \mu)$ is a Banach space.

Exercise 2.11. Let $1 \le p < p' < \infty$. Show that $\ell_p \subset \ell_{p'}$.

Solution. Let $x \in \ell_p$. The series $\sum_j |x_j|^p$ converges, so $|x_j| \to 0$, and there is an N with $|x_j| < 1$ for all $j > N$. For such j the larger exponent gives the smaller value, $|x_j|^{p'} \le |x_j|^p$, whence

$$\sum_{j>N} |x_j|^{p'} \le \sum_{j>N} |x_j|^p < \infty.$$

The remaining finitely many terms are finite, so $\sum_j |x_j|^{p'} < \infty$, that is $x \in \ell_{p'}$. In particular $\ell_1 \subset \ell_{3/2} \subset \ell_2 \subset \cdots \subset \ell_\infty$. ◊

Exercise 2.12. Let $1 \le p < p' < \infty$. Show that $L_p[0,1] \supset L_{p'}[0,1]$, the inclusion running the other way than in Exercise 2.11.

Solution. Work with the representatives of Remark 2.9, so that $f \in L_{p'}[0,1]$ means $\int_0^1 |f|^{p'} < \infty$. Split the domain according to the size of $|f|$:

$$\int_0^1 |f|^p = \int_{\{|f|>1\}} |f|^p + \int_{\{|f|\le 1\}} |f|^p \le \int_0^1 |f|^{p'} + 1 < \infty.$$

On the first set $|f|^p \le |f|^{p'}$, since $p < p'$ and $|f| > 1$ there; the second integral is at most the measure of $[0,1]$, which is 1. Hence $f \in L_p[0,1]$. ◊

3. Equivalent norms

Definition 3.1 (Equivalent norms)**.** Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space X are *equivalent* if there are constants $c, C > 0$ such that $c\|x\|_a \le \|x\|_b \le C\|x\|_a$ for all $x \in X$.

Equivalent norms induce the same topology, hence the same convergent sequences with the same limits, and the same Cauchy sequences. As a rule of thumb, the norms occurring naturally on an infinite-dimensional space are not equivalent; the finite-dimensional case is the opposite extreme.

Lemma 3.2. Let $(X, \|\cdot\|)$ be a normed space over $\mathbb{K}$ and let $x_1, \dots, x_n \in X$ be linearly independent. Then there is a $C > 0$ such that

$$\Big\|\sum_{i=1}^n \alpha_i x_i\Big\| \;\ge\; C \sum_{i=1}^n |\alpha_i| \qquad \text{for all } \alpha_1, \dots, \alpha_n \in \mathbb{K}.$$

Proof. Both sides are positively homogeneous of degree one in $(\alpha_1, \dots, \alpha_n)$, and both vanish when every α_i is zero, so it suffices to find a $C > 0$ that works for all coefficient vectors with $\sum_i |\alpha_i| = 1$.

Suppose no such C exists. Then for every $m \in \mathbb{N}$ there are coefficients $\alpha_1^{(m)}, \dots, \alpha_n^{(m)}$ with $\sum_i |\alpha_i^{(m)}| = 1$ and

$$\Big\|\sum_{i=1}^n \alpha_i^{(m)} x_i\Big\| < \frac{1}{m}.$$

Each coordinate sequence $(\alpha_i^{(m)})_m$ is bounded by 1, so by the Bolzano–Weierstrass theorem it has a convergent subsequence. Extracting successively for $i = 1, \dots, n$ and passing to the last subsequence, we may assume $\alpha_i^{(m)} \to \alpha_i$ for every i simultaneously. Then $\sum_i |\alpha_i| = \lim_m \sum_i |\alpha_i^{(m)}| = 1$, so not every α_i is zero. The map $(\alpha_1, \dots, \alpha_n) \mapsto \sum_i \alpha_i x_i$ is continuous, and so is the norm, hence

$$\Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| = \lim_{m \to \infty} \Big\| \sum_{i=1}^{n} \alpha_i^{(m)} x_i \Big\| = 0,$$

so $\sum_i \alpha_i x_i = 0$ with coefficients not all zero. This contradicts the independence of $x_1, \dots, x_n$. ∎

Theorem 3.3. If X is a finite-dimensional vector space over $\mathbb{K}$, then any two norms on X are equivalent.

Proof. Let $x_1, \dots, x_n$ be a basis of X and, for $x = \sum_{i=1}^n \alpha_i x_i$, set $\|x\|_1 = \sum_{i=1}^n |\alpha_i|$, which is a norm on X. Let $\|\cdot\|$ be any norm on X. By Lemma 3.2 there is a $C > 0$ with $\|x\| \geq C\|x\|_1$ for all $x \in X$. In the other direction,

$$\|x\| = \Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| \leq \sum_{i=1}^{n} |\alpha_i| \|x_i\| \leq \Big(\max_{1 \leq i \leq n} \|x_i\| \Big) \sum_{i=1}^{n} |\alpha_i| = c' \|x\|_1,$$

with $c' = \max_{1 \leq i \leq n} \|x_i\| > 0$, the maximum being positive because the basis vectors are non-zero. So every norm on X is equivalent to $\|\cdot\|_1$.

Equivalence of norms is transitive: if $c_a \|x\|_1 \leq \|x\|_a \leq C_a \|x\|_1$ and $c_b \|x\|_1 \leq \|x\|_b \leq C_b \|x\|_1$ for all x, then

$$\frac{c_b}{C_a} \|x\|_a \leq \|x\|_b \leq \frac{C_b}{c_a} \|x\|_a.$$

Applying this to two arbitrary norms on X, each equivalent to $\|\cdot\|_1$ by the first part, gives the claim. ∎

4. Separable spaces

Definition 4.1 (Countable set)**.** A set S is *countable* if it is finite or can be listed as $S = \{s_1, s_2, s_3, \dots\}$.

Definition 4.2 (Dense set)**.** A set $S \subset (X, d)$ is *dense* if $\overline{S} = X$.

Definition 4.3 (Separable space)**.** A metric space (X, d) is *separable* if it contains a countable dense set.

Remark 4.4. $S \subset (X, d)$ is dense if and only if for every $x \in X$ there is a sequence $(s_n) \subset S$ with $s_n \to x$.

The spaces introduced so far are now checked for completeness and for separability.

Example 4.5. $(\mathbb{R}, |\cdot|)$ is Banach and separable, with $\mathbb{Q}$ as a countable dense set.

Example 4.6. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then $(\mathbb{K}^n, \|\cdot\|_2)$ is Banach and separable: $\mathbb{Q}^n$ is a countable dense set for $\mathbb{K} = \mathbb{R}$, and $(\mathbb{Q} + i\mathbb{Q})^n$ for $\mathbb{K} = \mathbb{C}$.

Example 4.7. $(\mathbb{K}^n, \|\cdot\|_p)$ for $1 \le p < \infty$ and $(\mathbb{K}^n, \|\cdot\|_\infty)$ are Banach and separable, because these norms are equivalent to $\|\cdot\|_2$ by Theorem 3.3, and equivalent norms have the same convergent sequences and the same dense sets.

Remark 4.8. On a finite-dimensional space any such argument may be carried out in the Euclidean norm, since all norms there are equivalent.

Example 4.9. $(\ell_p, \|\cdot\|_p)$ is separable for $1 \le p < \infty$. The countable dense set consists of the finitely supported rational sequences $q = (q_1, \dots, q_m, 0, 0, \dots)$ with $q_i \in \mathbb{Q}$, or with $q_i \in \mathbb{Q} + i\mathbb{Q}$ when $\mathbb{K} = \mathbb{C}$: an $x = (x_1, x_2, \dots) \in \ell_p$ is approximated by truncating it and then approximating the first m coordinates.

Theorem 4.10. $(\ell_p, \|\cdot\|_p)$ is a Banach space for $1 \le p < \infty$.

Proof. Let $(x_m) \subset \ell_p$ be Cauchy, $x_m = (\xi_{1,m}, \xi_{2,m}, \dots)$. Given $\varepsilon > 0$ there is an N such that

$$\Big(\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_{j,n}|^p\Big)^{1/p} \le \varepsilon \qquad \text{for all } n, m > N.$$

A single term is at most the whole sum, so $|\xi_{j,m} - \xi_{j,n}| \le \varepsilon$ for every j and all $n, m > N$. Thus each coordinate sequence $(\xi_{j,m})_m$ is Cauchy in $\mathbb{K}$ and converges; write $\xi_j = \lim_{m\to\infty} \xi_{j,m}$ and $x = (\xi_1, \xi_2, \dots)$.

Fix k. For all $n, m > N$ the partial sum satisfies $\sum_{j=1}^{k} |\xi_{j,m} - \xi_{j,n}|^p \le \varepsilon^p$. Fix $m > N$ and let $n \to \infty$; the sum is finite, so the limit may be taken termwise, and

$$\sum_{j=1}^{k} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p.$$

The bound does not depend on k, so letting $k \to \infty$ gives $\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p$, that is $x_m - x \in \ell_p$ and $\|x_m - x\|_p \le \varepsilon$ for every $m > N$.

Consequently $x = (x - x_m) + x_m$ lies in ℓ_p, by Theorem 2.7 and $x_m \in \ell_p$. The estimate $\|x_m - x\|_p \le \varepsilon$ for all $m > N$ is convergence in the norm, not merely coordinatewise, so $x_m \to x$ in ℓ_p. ∎

Counterexample 4.11. $(C[0,1], \|\cdot\|_p)$ is not complete, for $1 \le p < \infty$.

Discussion. A first attempt takes $f_n(x) = x^n$. These are continuous, and

$$\|f_n - 0\|_p = \Big(\int_0^1 x^{np} \mathrm{d}x\Big)^{1/p} = \Big(\frac{1}{np+1}\Big)^{1/p} \longrightarrow 0,$$

so (f_n) is Cauchy but converges in $\|\cdot\|_p$ to the function 0, which is continuous. The attempt establishes nothing: a Cauchy sequence whose limit is continuous is no obstruction to completeness.

Take instead the ramps

$$f_n(x) = \begin{cases} 0, & 0 \le x \le \frac{1}{2} - \frac{1}{n}, \\ n\big(x - \frac{1}{2} + \frac{1}{n}\big), & \frac{1}{2} - \frac{1}{n} \le x \le \frac{1}{2}, \\ 1, & \frac{1}{2} \le x \le 1, \end{cases} \qquad n \ge 3.$$

For $n < m$ the functions f_n and f_m agree outside $[\frac{1}{2} - \frac{1}{n}, \frac{1}{2}]$ and differ by at most 1 on it, so $\|f_n - f_m\|_p \le n^{-1/p}$ and (f_n) is Cauchy in $\|\cdot\|_p$.

Suppose $g \in C[0,1]$ satisfied $\|f_n - g\|_p \to 0$. Fix $\delta > 0$. For $n > 1/\delta$ the function f_n vanishes on $[0, \frac{1}{2} - \delta]$, so

$$\int_0^{1/2-\delta} |g|^p \le \|f_n - g\|_p^p \longrightarrow 0,$$

whence $g \equiv 0$ on $[0, \frac{1}{2} - \delta]$, and since $\delta > 0$ was arbitrary, $g \equiv 0$ on $[0, \frac{1}{2})$; continuity gives $g(\frac{1}{2}) = 0$. The same estimate on $[\frac{1}{2}, 1]$, where $f_n \equiv 1$, gives $g \equiv 1$ there, so $g(\frac{1}{2}) = 1$. This is a contradiction, so no continuous limit exists. ◊

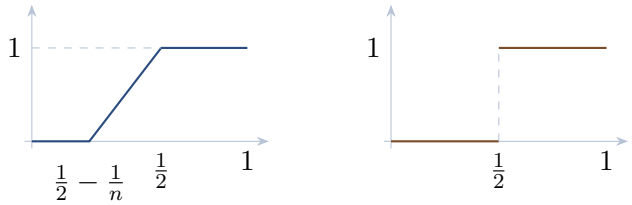


Figure 4.1. *The ramps f_n and the step function they approach. The step function is not continuous, so the limit leaves $C[0,1]$.*

Example 4.12. $L_p[0,1]$ and $L_p(\mathbb{R})$ are Banach and separable for $1 \le p < \infty$. For $L_p[0,1]$ either description may be used: the completion of $(C[0,1], \|\cdot\|_p)$, or the Lebesgue measurable f with $(\int |f|^p)^{1/p} < \infty$. Separability comes from the polynomials with rational coefficients, $p(x) = q_m x^m + \cdots + q_1 x + q_0$ with $q_i \in \mathbb{Q}$, which are dense by Bernstein's theorem.

Theorem 4.13. $(C[0,1], \|\cdot\|_\infty)$ is a Banach space.

Proof. Let $(f_n) \subset C[0,1]$ be a Cauchy sequence. Given $\varepsilon > 0$ there is an N such that $\|f_n - f_m\|_\infty = \max_{x\in[0,1]} |f_n(x) - f_m(x)| < \varepsilon$ for all $n, m > N$, and hence $|f_n(x) - f_m(x)| < \varepsilon$ for every x. So for fixed $x \in [0,1]$ the sequence $(f_n(x))$ is Cauchy in $\mathbb{K}$ and converges; write $f(x) = \lim_{n\to\infty} f_n(x)$. It remains to show that f is continuous and that $f_n \to f$ uniformly.

Let $n > N$. For all $m > N$ one has $|f_n(x) - f_m(x)| < \varepsilon$; letting $m \to \infty$ gives $|f_n(x) - f(x)| \le \varepsilon$ for all $n > N$ and all $x \in [0,1]$, which is uniform convergence.

Now fix $x \in [0,1]$ and $\varepsilon > 0$, and choose N with $|f_N(z) - f(z)| < \frac{\varepsilon}{3}$ for all $z \in [0,1]$. The function f_N is continuous on the compact interval $[0,1]$, hence uniformly continuous, so there is a $\delta > 0$ with $|f_N(x) - f_N(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta$. For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

so $f \in C[0,1]$. ■

Example 4.14. $(C[0,1], \|\cdot\|_\infty)$ is also separable: the separability argument of Example 4.12, with polynomials with rational coefficients, applies here as well.

Example 4.15. Let $C_p^r[0,1]$ carry $\|f\| := \|f\|_p + \|f'\|_p + \cdots + \|f^{(r)}\|_p$. This is a norm, but the space is not complete. Its completion is the Sobolev space $L_p^r[0,1]$.

5. Compact sets

Definition 5.1 (Compactness)**.** A set $K \subset (X,d)$ is *compact* if every open covering of K admits a finite subcovering.

Theorem 5.2. For $K \subset (X,d)$ the following are equivalent.

1. K is compact.
2. Every sequence in K has a convergent subsequence whose limit lies in K.
3. K is complete and totally bounded, that is, for every $\varepsilon > 0$ there are finitely many balls $B(x_1,\varepsilon),\dots,B(x_N,\varepsilon)$ with $K \subset \bigcup_{i=1}^{N} B(x_i,\varepsilon)$.

Lemma 5.3. A compact $K \subset (X,\|\cdot\|)$ is closed and bounded.

Proof. For closedness, let $(x_n) \subset K$ with $x_n \to x$. By Theorem 5.2 the sequence has a subsequence converging to a point of K, and that limit is x, so $x \in K$.

For boundedness, suppose K were unbounded. Then there is a sequence $(x_n) \subset K$ with $\|x_n\| \geq n$, which has no convergent subsequence, contradicting Theorem 5.2. ∎

Theorem 5.4 (Heine–Borel)**.** A set $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Definition 5.5 (Local compactness)**.** A metric space (X,d) is *locally compact* if every $x \in X$ has a neighbourhood $B(x,\varepsilon)$ whose closure $\overline{B(x,\varepsilon)}$ is compact.

Definition 5.6 (Relative compactness)**.** A set $S \subset (X,d)$ is *relatively compact* if $\overline{S}$ is compact.

Lemma 5.7 (Riesz)**.** Let $M \subset (X,\|\cdot\|)$ be a proper closed subspace and let $t \in (0,1)$. Then there is an $x_t \in X$ with $\|x_t\| = 1$ and $\operatorname{dist}(x_t, M) \geq t$, where $\operatorname{dist}(x,M) = \inf\{\|x-m\| : m \in M\}$.

Proof. Since M is a proper subspace there is a $u \in X \setminus M$, and since M is closed, $d := \operatorname{dist}(u,M) > 0$. As $t \in (0,1)$ we have $d/t > d$, so by the definition of the infimum there is an $x_0 \in M$ with

$$d \leq \|u - x_0\| \leq \frac{d}{t}.$$

Put $x_t = (u-x_0)/\|u-x_0\|$, a unit vector. Let $x \in M$ be arbitrary and set $x_1 = \|u-x_0\|\,x + x_0$, which lies in M because M is a subspace. Then

$$\|x - x_t\| = \frac{1}{\|u-x_0\|}\big\|\|u-x_0\|\,x - u + x_0\big\| = \frac{\|x_1 - u\|}{\|u-x_0\|} \geq \frac{d}{\|u-x_0\|} \geq \frac{d}{d/t} = t,$$

using $\|x_1 - u\| \geq \operatorname{dist}(u,M) = d$. As $x \in M$ was arbitrary, $\operatorname{dist}(x_t, M) \geq t$. ∎

Theorem 5.8. A normed space $(X, \|\cdot\|)$ is locally compact if and only if $\dim X < \infty$.

Proof. For $x \in X$ and $\varepsilon > 0$ the map $y \mapsto x + \varepsilon y$ is a homeomorphism of X carrying $\overline{B(0,1)}$ onto $\overline{B(x,\varepsilon)}$. All closed balls are therefore compact together with $\overline{B(0,1)}$, and X is locally compact if and only if $\overline{B(0,1)}$ is compact.

Assume $\dim X = \infty$. Choose a unit vector $x_1 \in X$ and put $M_1 = \operatorname{span}\{x_1\}$. Each $M_i = \operatorname{span}\{x_1, \dots, x_i\}$ is finite-dimensional, hence closed: by Theorem 3.3 its norm is equivalent to the norm transported from $\mathbb{K}^i$, which is complete, so M_i is complete and therefore closed in X. It is proper because $\dim X = \infty$. So Lemma 5.7 with $t = \frac{1}{2}$ yields a unit vector x_{i+1} with $\operatorname{dist}(x_{i+1}, M_i) \geq \frac{1}{2}$. This produces a sequence of unit vectors with

$$\|x_n - x_m\| \geq \tfrac{1}{2} \qquad \text{for } n \neq m,$$

since for $n > m$ one has $x_m \in M_{n-1}$. The sequence lies in $\overline{B(0,1)}$ and has no convergent subsequence, so $\overline{B(0,1)}$ is not compact by Theorem 5.2, and X is not locally compact.

Assume $\dim X = n < \infty$. Fixing a basis identifies X with $\mathbb{K}^n$, and by Theorem 3.3 the norm of X is equivalent to the Euclidean norm transported by that identification, so the two have the same closed sets and the same bounded sets. The set $\overline{B(0,1)}$ is closed and bounded, hence compact by Theorem 5.4, and X is locally compact. ∎

Corollary 5.9. $\overline{B(0,1)} \subset (X, \|\cdot\|)$ is compact if and only if $\dim X < \infty$.

6. Compact sets in $(C[a,b], \|\cdot\|_\infty)$

Definition 6.1 (Uniform boundedness)**.** A set $S \subset C[a,b]$ is *uniformly bounded* if there is a $K > 0$ with $\|f\|_\infty \leq K$ for every $f \in S$.

Definition 6.2 (Uniform equicontinuity)**.** A set $S \subset C[a,b]$ is *uniformly equicontinuous* if for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every $f \in S$ and all $x, y \in [a,b]$ with $|x - y| < \delta$ one has $|f(x) - f(y)| < \varepsilon$.

Theorem 6.3 (Arzelà–Ascoli)**.** A set $S \subset (C[a,b], \|\cdot\|_\infty)$ is relatively compact if and only if it is uniformly bounded and uniformly equicontinuous. If S is in addition closed, it is compact.

Proof. Assume first that S is relatively compact. Then $\overline{S}$ is compact, hence bounded by Lemma 5.3, so S is uniformly bounded. Fix $\varepsilon > 0$. By Theorem 5.2 the compact set $\overline{S}$ is totally bounded, so there are finitely many $f_1, \dots, f_r$ with $\overline{S} \subset \bigcup_{i=1}^{r} B(f_i, \frac{\varepsilon}{3})$. Each f_i is continuous on the compact interval $[a,b]$, hence uniformly continuous, so there is a $\delta_i > 0$ with $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta_i$. Put $\delta = \min_{1 \leq i \leq r} \delta_i$, which is positive because the minimum is over finitely many indices. Given $f \in S$, choose i with $\|f - f_i\|_\infty < \frac{\varepsilon}{3}$. Then for $|x - y| < \delta$,

$$|f(x) - f(y)| \leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < \varepsilon,$$

so S is uniformly equicontinuous.

Assume now that S is uniformly bounded, by K, and uniformly equicontinuous, and let $(f_n) \subset S$. The goal is to select a convergent subsequence. Enumerate the rationals of the interval,

$\mathbb{Q} \cap [a,b] = \{x_1, x_2, \dots\}$. The values $f_n(x_1)$ satisfy $|f_n(x_1)| \le K$, so some subsequence $(f_n^{(1)})$ converges at x_1. Along it, the values at x_2 are again bounded, so a further subsequence $(f_n^{(2)})$ converges at x_2, and so on. Let $g_m = f_m^{(m)}$ be the diagonal sequence. It is a subsequence of the original (f_n), and from the index k onwards it is a subsequence of $(f_n^{(k)})$, so $(g_m(x_k))$ converges for every k.

The claim is that (g_m) is uniformly Cauchy. Fix $\varepsilon > 0$ and let $\delta > 0$ be given by uniform equicontinuity for $\frac{\varepsilon}{3}$. Cover $[a,b]$ by finitely many sets $U_1, \dots, U_s$ of diameter less than δ, and pick in each U_j a rational point x_{k_j}, which exists because the rationals are dense. For each j the sequence $(g_m(x_{k_j}))_m$ converges, hence is Cauchy, so there is an N_j with $|g_n(x_{k_j}) - g_m(x_{k_j})| < \frac{\varepsilon}{3}$ for $n, m > N_j$. Put $N = \max_{1 \le j \le s} N_j$, finite because there are finitely many j. Let $t \in [a,b]$ and choose j with $t \in U_j$; then $|t - x_{k_j}| < \delta$, so for $n, m > N$,

$$|g_n(t) - g_m(t)| \le |g_n(t) - g_n(x_{k_j})| + |g_n(x_{k_j}) - g_m(x_{k_j})| + |g_m(x_{k_j}) - g_m(t)| < \varepsilon.$$

Hence $\|g_n - g_m\|_\infty \le \varepsilon$ for $n, m > N$. The space $(C[a,b], \|\cdot\|_\infty)$ is complete, by the proof of Theorem 4.13 with $[a,b]$ in place of $[0,1]$, so (g_m) converges in it.

Finally, $\overline{S}$ has the same uniform bound K and the same modulus of equicontinuity as S, so the selection above applies verbatim to a sequence in $\overline{S}$, and its limit lies in $\overline{S}$ because $\overline{S}$ is closed. By Theorem 5.2 the set $\overline{S}$ is compact, that is, S is relatively compact. ∎

7. Bases

Definition 7.1 (Algebraic basis)**.** Let X be a vector space. A set $S \subset X$ is an *algebraic basis*, or *Hamel basis*, if S is linearly independent and every $x \in X$ has a representation $x = \sum_{i \in I} \alpha_i s_i$ with $s_i \in S$, $\alpha_i \in \mathbb{K}$ and I finite.

Example 7.2. Let $1 \le p < \infty$ and $e_n = (0, \dots, 0, 1, 0, \dots)$, with the 1 in position n. The set $E = \{e_1, e_2, \dots\}$ is not an algebraic basis of ℓ_p: a finite combination of the e_n has only finitely many non-zero coordinates, while $(2^{-n})_n \in \ell_p$ has infinitely many.

Definition 7.3 (Convergent series)**.** In a normed space X a series $\sum_{n=1}^{\infty} x_n$ *converges* to $x \in X$ if its partial sums $s_N := \sum_{n=1}^{N} x_n$ satisfy $s_N \to x$.

Definition 7.4 (Schauder basis)**.** In a Banach space X a sequence (s_n) is a *Schauder basis* if every $x \in X$ has a unique representation $x = \sum_{i=1}^{\infty} \alpha_i s_i$ with $\alpha_i \in \mathbb{K}$, the series converging in the sense of Definition 7.3.

Example 7.5. $E = (e_n)$ is a Schauder basis of ℓ_p for $1 \le p < \infty$.

Proof. Let $x = (\alpha_1, \alpha_2, \dots) \in \ell_p$ and $s_N = \sum_{i=1}^{N} \alpha_i e_i = (\alpha_1, \dots, \alpha_N, 0, \dots)$. Then

$$\|x - s_N\|_p^p = \|(0, \dots, 0, \alpha_{N+1}, \alpha_{N+2}, \dots)\|_p^p = \sum_{i=N+1}^{\infty} |\alpha_i|^p \longrightarrow 0,$$

being the tail of a convergent series, so $x = \sum_i \alpha_i e_i$. For uniqueness, suppose $x = \sum_i \beta_i e_i$. The j-th coordinate of the N-th partial sum is β_j once $N \ge j$, and a coordinate satisfies $|y_j| \le \|y\|_p$, so coordinates pass to the limit and $\beta_j = \alpha_j$. ∎

Remark 7.6. E is not a Schauder basis of ℓ_∞. Take $x = (1, 1, \dots)$. Since $|y_j| \le \|y\|_\infty$, the argument above shows that any representation $x = \sum_i \beta_i e_i$ must have $\beta_i = 1$ for all i. But then

$$\|x - s_N\|_\infty = \|(0, \dots, 0, 1, 1, \dots)\|_\infty = 1$$

for every N, so the partial sums do not converge to x, and x has no representation at all.

Remark 7.7. ℓ_∞ is not separable. The sequences with every term 0 or 1 form an uncountable set, and any two distinct ones have $\|\cdot\|_\infty$ distance 1. The balls of radius $\frac{1}{3}$ around them are therefore pairwise disjoint: a common point would put the two centres at distance at most $\frac{2}{3}$. A dense set meets every one of these balls, and meets distinct balls in distinct points, so it is uncountable.

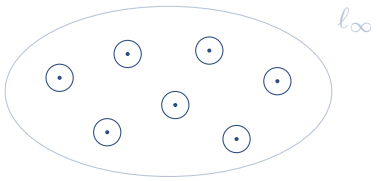


Figure 7.1. *Around the uncountably many* 0–1 *sequences, the balls of radius* $\frac{1}{3}$ *in* ℓ_∞ *are pairwise disjoint, and a dense set needs a point in each.*

Example 7.8. The Fourier systems are Schauder bases. In the real space $L_2[0, 2\pi]$ the system $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ is one: every f has a unique representation

$$f = a_0 + \sum_{k=1}^{\infty} \big(a_k \cos(kx) + b_k \sin(kx)\big).$$

In the complex space $L_2[0, 2\pi]$ the system e^{ikx}, $k \in \mathbb{Z}$, enumerated as $1, e^{ix}, e^{-ix}, e^{2ix}, e^{-2ix}, \dots$, is one: every f has a unique representation $f = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$, meaning

$$\Big\| f - \sum_{k=-N}^{N} c_k e^{ikx} \Big\|_2^2 = \int_0^{2\pi} \Big| f(x) - \sum_{k=-N}^{N} c_k e^{ikx} \Big|^2 \mathrm{d}x \xrightarrow[N\to\infty]{} 0.$$

Convergence in $\|\cdot\|_2$ does not guarantee pointwise convergence.

Proposition 7.9. A Banach space with a Schauder basis is separable.

Proof. Let (s_i) be a Schauder basis and let D be the set of finite sums $\sum_{i=1}^{N} q_i s_i$ with $N \in \mathbb{N}$ and $q_i \in \mathbb{Q}$, or $q_i \in \mathbb{Q} + i\mathbb{Q}$ when $\mathbb{K} = \mathbb{C}$. For each N these sums form a countable set, so D is countable. Given $x = \sum_i \alpha_i s_i$ and $\varepsilon > 0$, choose N with $\|x - \sum_{i=1}^{N} \alpha_i s_i\| < \frac{\varepsilon}{2}$, and then rational q_i with $|\alpha_i - q_i| \, \|s_i\| < \frac{\varepsilon}{2N}$ for $i \le N$. The triangle inequality gives $\|x - \sum_{i=1}^{N} q_i s_i\| < \varepsilon$, so D is dense. ∎

Corollary 7.10. ℓ_∞ has no Schauder basis.

Proof. It is not separable, by Remark 7.7, so Proposition 7.9 excludes a Schauder basis. ∎

Lemma 7.11 (Zorn)**.** Let $(P, \leq)$ be a partially ordered set in which every chain, that is every totally ordered subset, has an upper bound in P. Then P has a maximal element, an $m \in P$ such that $m \leq p$ implies $p = m$.

Remark 7.12. The *axiom of choice* states that $\prod_{i \in I} A_i$ is non-empty whenever every A_i is non-empty, that is, that there is a family $(a_i)_{i \in I}$ with $a_i \in A_i$ for every i.

Theorem 7.13. Every vector space has an algebraic basis.

Remark 7.14. An algebraic basis has no connection to any norm on the space.

Proof of Theorem 7.13. Let P be the family of linearly independent subsets of the vector space X, and write $I \leq J$ if $I \subseteq J$. This is a partial order: it is reflexive, $I \leq I$; antisymmetric, $I \leq J$ and $J \leq I$ give $I = J$; and transitive, $I \leq J$ and $J \leq K$ give $I \leq K$. The family is non-empty, since $\emptyset \in P$.

Let $\mathcal{C} \subset P$ be a chain. Its union U is linearly independent: finitely many $u_1, \ldots, u_k \in U$ lie in members $C_1, \ldots, C_k$ of $\mathcal{C}$, which are totally ordered, so the largest of them contains all u_j, and these are therefore independent. Thus $U \in P$ is an upper bound of $\mathcal{C}$; for the empty chain, $\emptyset$ is one. By Lemma 7.11 there is a maximal independent set S.

S spans X. Otherwise there is an $x \notin \operatorname{span} S$, and $S \cup \{x\}$ is independent: in a vanishing combination the coefficient of x must be 0, since otherwise $x \in \operatorname{span} S$, and then the remaining coefficients vanish by independence of S. This contradicts maximality, so S is an algebraic basis. ∎

Definition 7.15 (Absolute convergence)**.** A series $\sum_{i=1}^{\infty} x_i$ in a normed space X is *absolutely convergent* if $\sum_{i=1}^{\infty} \|x_i\| < \infty$.

Lemma 7.16 (Absolute convergence)**.** If X is a Banach space and $\sum_{i=1}^{\infty} x_i$ is absolutely convergent, then $\sum_{i=1}^{\infty} x_i$ converges.

Proof. As X is complete, it suffices to show that the partial sums $s_N = \sum_{i=1}^{N} x_i$ form a Cauchy sequence. Let $\varepsilon > 0$. The real series $\sum_i \|x_i\|$ converges, so there is an N with $\sum_{i=m+1}^{n} \|x_i\| < \varepsilon$ for all $n > m > N$. For such n, m,

$$\|s_n - s_m\| = \Big\| \sum_{i=m+1}^{n} x_i \Big\| \leq \sum_{i=m+1}^{n} \|x_i\| < \varepsilon. \qquad ∎$$

8. Bounded linear operators

Definition 8.1 (Linear operator)**.** Let X and Y be vector spaces over $\mathbb{K}$. A map $T \colon X \to Y$ is *linear* if $T(x_1 + x_2) = Tx_1 + Tx_2$ for all $x_1, x_2 \in X$ and $T(\alpha x) = \alpha\, Tx$ for all $\alpha \in \mathbb{K}$ and $x \in X$.

Definition 8.2 (Bounded operator, operator norm)**.** A linear operator $T\colon (X, \|\cdot\|) \to (Y, \|\cdot\|)$ between normed spaces is *bounded* if $\sup_{\|x\|\le 1}\|Tx\| < \infty$. In that case

$$\|T\| = \sup_{\|x\|\le 1} \|Tx\|$$

is the *operator norm* of T.

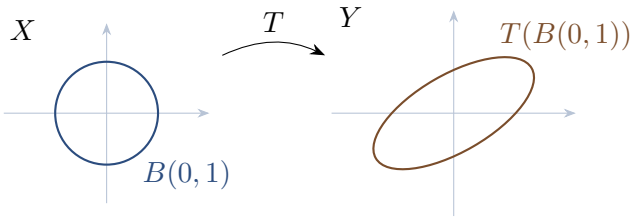


Figure 8.1. *A bounded operator maps the closed unit ball to a bounded set, here an ellipse; $\|T\|$ is the radius of the smallest ball about 0 containing the image.*

Remark 8.3. For every $x \in X$, $\|Tx\| \le \|T\|\,\|x\|$.

Proof. For $x = 0$ both sides vanish, since $T0 = 0$ by linearity. For $x \neq 0$, linearity gives $Tx = \|x\|\,T\big(x/\|x\|\big)$, and $x/\|x\|$ has norm 1, so

$$\|Tx\| = \|x\|\,\Big\|T\Big(\frac{x}{\|x\|}\Big)\Big\| \le \|x\| \sup_{\|x'\|\le 1}\|Tx'\| = \|x\|\,\|T\|. \qquad \blacksquare$$

Example 8.4. Every linear operator $T\colon X \to Y$ on a finite-dimensional normed space X is bounded. In particular every linear $T\colon \mathbb{K}^n \to \mathbb{K}^m$, which is given by an $m \times n$ matrix, is bounded.

Proof. Let $e_1, \dots, e_n$ be a basis of X. By Lemma 3.2 there is a $C > 0$ with $\|\sum_i \alpha_i e_i\| \ge C \sum_i |\alpha_i|$ for all coefficients. For $x = \sum_{i=1}^n \alpha_i e_i$,

$$\|Tx\| = \Big\|\sum_{i=1}^n \alpha_i\, Te_i\Big\| \le \sum_{i=1}^n |\alpha_i|\,\|Te_i\| \le \Big(\max_{1\le i\le n}\|Te_i\|\Big)\sum_{i=1}^n |\alpha_i| \le \frac{1}{C}\max_{1\le i\le n}\|Te_i\|\;\|x\|,$$

so $\|T\| \le C^{-1}\max_i\|Te_i\| < \infty$. $\blacksquare$

Example 8.5. For fixed $x \in \mathbb{K}^n$ let $f_x\colon \mathbb{K}^n \to \mathbb{K}$, $f_x(y) = \sum_{i=1}^n x_i y_i$. Then f_x is linear, and bounded by Example 8.4 with $m = 1$.

Definition 8.6 (Linear functional)**.** A linear operator from X to $\mathbb{K}$ is a *linear functional*.

Example 8.7 (Multiplication operator on $C[0,1]$)**.** Fix $f \in C[0,1]$ and let $M_f\colon C[0,1] \to C[0,1]$, $g \mapsto fg$, with the norm $\|\cdot\|_\infty$. Then M_f is linear and bounded, with $\|M_f\| \le \|f\|_\infty$.

Proof. $\|f\|_\infty < \infty$ because f is continuous on the compact $[0,1]$. For $\|g\|_\infty \le 1$,

$$\|M_f g\|_\infty = \sup_{x\in[0,1]} |f(x)g(x)| \le \|f\|_\infty\|g\|_\infty \le \|f\|_\infty. \qquad \blacksquare$$

Example 8.8 (Multiplication operator on ℓ_p)**.** Let $1 \le p < \infty$, fix $x = (x_1, x_2, \dots) \in \ell_\infty$, and let $M_x \colon \ell_p \to \ell_p$, $y = (y_1, y_2, \dots) \mapsto (x_1 y_1, x_2 y_2, \dots)$. Then M_x is linear and bounded, with $\|M_x\| \le \|x\|_\infty$.

Proof. Linearity is immediate. Termwise $|x_i y_i|^p \le \|x\|_\infty^p |y_i|^p$, and summing gives $\|M_x y\|_p \le \|x\|_\infty \|y\|_p$. This shows first that $M_x y \in \ell_p$, so that M_x is well defined, and then that $\|M_x y\|_p \le \|x\|_\infty$ whenever $\|y\|_p \le 1$. ∎

Remark 8.9. Multiplication operators are the simplest operators. They are the counterpart of diagonal matrices: $D = \operatorname{diag}(\alpha_1, \dots, \alpha_n)$ acts by $y \mapsto Dy = (\alpha_1 y_1, \dots, \alpha_n y_n)$, which is multiplication by $(\alpha_1, \dots, \alpha_n)$, and diagonalizing a matrix in linear algebra reduces it to such an operator.

Example 8.10 (Integral functional)**.** Let $I \colon C[0,1] \to \mathbb{K}$, $I(f) = \int_0^1 f(t)\mathrm{d}t$, with the norm $\|\cdot\|_\infty$. Then I is linear and bounded, with $\|I\| \le 1$: for $\|f\|_\infty \le 1$,

$$|If| = \left| \int_0^1 f(t)\mathrm{d}t \right| \le \int_0^1 |f(t)|\mathrm{d}t \le \int_0^1 \mathrm{d}t = 1.$$

Example 8.11 (Evaluation functional)**.** Fix $t_0 \in [0,1]$ and let $E \colon C[0,1] \to \mathbb{K}$, $Ef = f(t_0)$, with the norm $\|\cdot\|_\infty$. Then E is linear and bounded, with $\|E\| \le 1$: for $\|f\|_\infty \le 1$, $|Ef| = |f(t_0)| \le \|f\|_\infty \le 1$.

Example 8.12 (Integral operator)**.** Fix a *kernel* $K \in C([0,1] \times [0,1])$ and let $T_K \colon C[0,1] \to C[0,1]$,

$$(T_K f)(x) = \int_0^1 K(x,y) f(y)\mathrm{d}y,$$

with the norm $\|\cdot\|_\infty$ on both sides. This is the continuous analogue of matrix multiplication, $(Kf)_j = \sum_{m=1}^n K_{jm} f_m$. Then T_K is linear and bounded, with $\|T_K\| \le \|K\|_\infty$, where $\|K\|_\infty = \max_{[0,1]^2} |K|$ is finite because K is continuous on a compact set.

Proof. $T_K f$ is continuous: K is uniformly continuous on the compact square, and $|(T_K f)(x) - (T_K f)(x')| \le \sup_y |K(x,y) - K(x',y)|\, \|f\|_\infty$. Linearity is immediate. For $\|f\|_\infty \le 1$,

$$\|T_K f\|_\infty = \sup_{x \in [0,1]} \left| \int_0^1 K(x,y) f(y)\mathrm{d}y \right| \le \sup_{x \in [0,1]} \int_0^1 |K(x,y)|\, |f(y)|\mathrm{d}y \le \|K\|_\infty \int_0^1 \mathrm{d}y = \|K\|_\infty. \quad ∎$$

Example 8.13 (Differentiation)**.** Let X be the space of polynomials on $[0,1]$ with the norm $\|\cdot\|_\infty$, a subspace of $C[0,1]$; it is not a Banach space. Let $D \colon X \to X$, $f \mapsto f'$. Then D is linear but not bounded: for $f_n(t) = t^n$ one has $\|f_n\|_\infty = 1$ for all n, while $\|Df_n\|_\infty = \|nt^{n-1}\|_\infty = n \to \infty$.

Remark 8.14. As a rule of thumb, integration is bounded and differentiation is not.

9. The space of bounded operators

Theorem 9.1. A linear operator $T\colon (X, \|\cdot\|) \to (Y, \|\cdot\|)$ is bounded if and only if it is continuous.

Proof. Assume T is bounded. If $T = 0$ it is continuous, so let $\|T\| > 0$. Given $\varepsilon > 0$ put $\delta = \varepsilon/\|T\|$. For all $x_1, x_2 \in X$ with $\|x_1 - x_2\| < \delta$, Remark 8.3 gives

$$\|Tx_1 - Tx_2\| = \|T(x_1 - x_2)\| \le \|T\|\,\|x_1 - x_2\| < \|T\|\frac{\varepsilon}{\|T\|} = \varepsilon.$$

Assume T is continuous; only continuity at 0 will be used. Since $T0 = 0$, the choice $\varepsilon = 1$ gives a $\delta > 0$ with $\|Tx\| < 1$ whenever $\|x\| < \delta$. For $\|z\| \le 1$ the vector $\frac{\delta}{2}z$ has norm less than δ, so

$$\|Tz\| = \frac{2}{\delta}\Big\|T\Big(\frac{\delta}{2}z\Big)\Big\| < \frac{2}{\delta}.$$

Hence T is bounded, with $\|T\| \le 2/\delta$. ∎

Corollary 9.2. A linear operator $T\colon (X, \|\cdot\|) \to (Y, \|\cdot\|)$ is continuous everywhere if and only if it is continuous at 0.

Proof. The second half of the proof of Theorem 9.1 used only continuity at 0 to obtain boundedness, and the first half turns boundedness into continuity everywhere. ∎

Definition 9.3. $\mathcal{B}(X, Y)$ denotes the set of all bounded linear operators from X to Y.

Theorem 9.4. $\mathcal{B}(X, Y)$ is a vector space under pointwise operations, and the operator norm is a norm on it. Therefore $(\mathcal{B}(X, Y), \|\cdot\|)$ is a normed space.

Proof. Pointwise sums and scalar multiples of linear maps are linear, and the zero operator is bounded. For $S, T \in \mathcal{B}(X, Y)$, $\alpha \in \mathbb{K}$ and $\|x\| \le 1$,

$$\|(S+T)x\| \le \|Sx\| + \|Tx\| \le \|S\| + \|T\|, \qquad \|(\alpha T)x\| = |\alpha|\,\|Tx\| \le |\alpha|\,\|T\|,$$

so $S + T$ and αT are bounded, and $\mathcal{B}(X, Y)$ is a subspace of the vector space of linear maps from X to Y.

For the norm axioms, $\|T\| \ge 0$ is clear. If $\|T\| = 0$, then Remark 8.3 gives $\|Tx\| \le \|T\|\,\|x\| = 0$ for every x, so $T = 0$. Homogeneity follows from $\|\alpha T\| = \sup_{\|x\|\le 1}|\alpha|\,\|Tx\| = |\alpha|\,\|T\|$. For the triangle inequality, the triangle inequality in Y gives

$$\|T + S\| = \sup_{\|x\|\le 1}\|(T+S)x\| \le \sup_{\|x\|\le 1}\big(\|Tx\| + \|Sx\|\big) \le \sup_{\|x\|\le 1}\|Tx\| + \sup_{\|x\|\le 1}\|Sx\| = \|T\| + \|S\|. \quad ∎$$

Theorem 9.5. If $(Y, \|\cdot\|)$ is a Banach space, then so is $(\mathcal{B}(X, Y), \|\cdot\|)$.

Corollary 9.6. $(\mathcal{B}(X, \mathbb{K}), \|\cdot\|)$ is a Banach space for every normed space X.

Notation 9.7. $X' = (\mathcal{B}(X, \mathbb{K}), \|\cdot\|)$ is the *dual space* of X.

Proof of Theorem 9.5. Let $(T_n) \subset \mathcal{B}(X, Y)$ be a Cauchy sequence: for every $\varepsilon > 0$ there is an N with $\|T_n - T_m\| \leq \varepsilon$ for all $n, m > N$. For each $x \in X$, Remark 8.3 gives

$$\|T_n x - T_m x\| = \|(T_n - T_m)x\| \leq \|T_n - T_m\| \, \|x\| \leq \varepsilon \|x\|,$$

and since ε is arbitrary, $(T_n x)$ is Cauchy in Y. As Y is complete, the pointwise limit $Tx := \lim_{n\to\infty} T_n x$ exists. It is linear, because $T(\alpha x + \beta y) = \lim_n (\alpha T_n x + \beta T_n y) = \alpha Tx + \beta Ty$.

It remains to show that T is bounded and that $T_n \to T$ in norm. Fix $n > N$ and let $m \to \infty$ in $\|T_n x - T_m x\| \leq \varepsilon \|x\|$; continuity of the norm gives

$$\|(T_n - T)x\| \leq \varepsilon \|x\| \qquad \text{for all } x \in X.$$

Thus $T_n - T$ is bounded with $\|T_n - T\| \leq \varepsilon$, so $T = T_n - (T_n - T) \in \mathcal{B}(X, Y)$ by Theorem 9.4, and $\|T_n - T\| \leq \varepsilon$ for all $n > N$ means $T_n \to T$. ∎

Proposition 9.8 (Composition)**.** If $T \in \mathcal{B}(Y, Z)$ and $S \in \mathcal{B}(X, Y)$, then $TS \in \mathcal{B}(X, Z)$ and $\|TS\| \leq \|T\| \, \|S\|$.

Proof. TS is linear as a composition of linear maps. By Remark 8.3,

$$\|TS\| = \sup_{\|x\|\leq 1} \|T(Sx)\| \leq \sup_{\|x\|\leq 1} \|T\| \, \|Sx\| = \|T\| \sup_{\|x\|\leq 1} \|Sx\| = \|T\| \, \|S\|. \qquad ∎$$

Remark 9.9. In particular, for a Banach space X the space $\mathcal{B}(X, X)$ is a Banach algebra: a Banach space, by Theorem 9.5, which is also an algebra under composition, with $\|TS\| \leq \|T\| \, \|S\|$.

Definition 9.10 (Invertible operator)**.** $T \in \mathcal{B}(X, X)$ is *invertible* if T is a bijection from X to X and T^{-1} is bounded.